

Javad Komijani

Curriculum Vitae

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Professional Summary

Highly analytical and detail-oriented researcher with an extensive background in mathematical methods, numerical simulations, theoretical modeling, statistical analysis, signal processing, and machine learning. Passionate about applying data science and machine learning to solve complex, real-world problems.

Education

- 2015 **PhD in Physics**, *Washington University in St. Louis*
Thesis: *Topics in Lattice Gauge Theory and Theoretical Physics*
- Developed theoretical models based on effective field theories
 - Applied Bayesian inference to lattice-QCD data
 - Estimated parameters of the Standard Model
 - Introduced new concepts in mathematical physics
- 2009 **MSc in Electrical Engineering**, *University of Tehran*
- Focus: electromagnetic waves and fields, signal processing, statistical analysis

Technical Skills

Programming & Tools	Python, Cython, C/C++, bash, MATLAB, Mathematica, PyTorch, Pandas, NumPy, SciPy, Scikit-learn, SQL, git
ML & AI	Generative Models, Diffusion Models, Normalizing Flows
Statistics	Bayesian inference, Monte Carlo simulations, statistical analysis, optimization
Computation	Numerical Simulations, High-Performance Computing (HPC)

Selected Projects

- Normflow** Normalizing Flows in PyTorch; supporting lattice gauge theory.
- torch_linagl** Linear Algebra Extension for PyTorch; supporting automatic differentiation for eigenvalue and singular value decompositions of unitary matrices.
- torch_solve** PyTorch Extension for Solving ODE; supporting automatic differentiation using the adjoint sensitivity method for unitary matrices.
- Diffusion** Diffusion Models in PyTorch; supporting unitary matrices.
- interTwin** Contributed the normflow package to the interTwin physics thematic module—integrated normalizing flows into the interTwin Digital Twin Engine for lattice QCD simulations.
- Quark Masses** Precisely calculated the up-, down-, strange-, charm-, and bottom-quark masses using the MILC highly improved staggered-quark ensembles.
- MRS Mass** Defined a novel quark mass scheme, the minimal-renormalon-subtracted (MRS) mass, which is free from the shortcomings of other quark masses and provides a more robust foundation for precise quark mass determination.
- Eigenvalues** Generalized eigenvalue problems to non-linear cases using properties of solutions to the Schrödinger equation.

Experience

2021–Present **Senior Postdoctoral Researcher, *ETH Zurich***

- Developed ML algorithms to generate lattice field configurations
- Worked extensively with HPC systems for lattice-QCD simulations
- Collaborated internationally on computational/theoretical projects
- Supervised PhD/MSc students and taught Quantum Mechanics

2019–2020 **Postdoctoral Researcher, *University of Tehran***

- Applied lattice-QCD simulations and Bayesian analysis to determine Standard Model parameters

2017–2018 **Postdoctoral Researcher, *University of Glasgow***

- Focused on determination of various form factors and quark condensates, using lattice-QCD Monte-Carlo simulations and Bayesian analysis

2015–2017 **Postdoctoral Researcher, *Technical University of Munich***

- Focused on determination of quark masses and various decay constants, using lattice-QCD Monte-Carlo simulations and Bayesian analysis

Language Skills

English Fluent

German Basic

Publications

A complete list of over 25 peer-reviewed publications can be found at [this link](#).